FEWELL

IOT HARDWARE & MOBILE OPERATIONAL MANUAL

Document Revision: 2.0
Target Platform: Android / ESP32

1 System Overview

FEWELL is an online-first Android telemetry and fuel monitoring platform engineered specifically for motorcycles. The system integrates an onboard ESP32 hardware microcontroller (measuring voltage and current dynamics across the fuel floater circuit via an INA219 sensor), real-time smartphone GPS tracking, local Room SQLite state management, and cloud database synchronization via Supabase.

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[Fuel Floater] → [INA219 Sensor] → [ESP32] --Bluetooth SPP--> [Android App] --Cellular/Wi-Fi--> [Supabase Cloud DB]
```

2 FEWELL IoT Installation Procedure

> **SAFETY REMINDER:** Verify the actual wiring diagram of your motorcycle's fuel floater and fuel gauge before connecting the INA219 sensor. The VIN+ and VIN− pins must be placed correctly in the current path to measure current accurately. **NEVER** connect a 12V accessory wire directly to ESP32 power pins that are not rated for that voltage (use an appropriate step-down buck converter).

Step 1: Remove the Motorcycle Seat and Rear Fairing

Remove the seat and rear body fairing to gain access to the motorcycle's internal wiring harness and electrical components.

Step 2: Remove Compartment and Fuel Tank Screws

Unscrew and remove the compartment and fuel tank mountings to expose the fuel floater socket and underlying wiring.

Step 3: Connect INA219 to Fuel Floater and Fuel Gauge

1. Unplug the fuel floater harness socket.
2. Use jumper or extension wires to connect the black wire from the original fuel floater socket directly to the fuel gauge.
3. Connect **VIN+** of the INA219 sensor module to the green wire coming from the fuel floater.
4. Connect **VIN−** of the INA219 sensor module to the green wire running toward the stock fuel gauge.
5. Inspect all electrical connections to ensure secure wiring and insulation.

Step 4: Connect the Power Supply of the FEWELL IoT

1. Connect the negative power wire (GND) of the IoT system to the negative (−) battery terminal.
2. Connect the positive power wire of the IoT system to the blue accessory wire located in the rear harness.
3. Verify the accessory wire voltage and install a step-down voltage regulator (12V to 5V) to safely power the ESP32, LCD, and sensor modules.

Step 5: Test the FEWELL IoT System

1. Inspect all wiring connections thoroughly before switching on main power.
2. Test the INA219 sensor to ensure it detects current flow when the ignition key is turned on.
3. Check the stock fuel gauge on the motorcycle dashboard to ensure it remains operational.
4. Test the ESP32, LCD display, and Bluetooth connection.
5. Confirm that fuel level telemetry data is successfully transmitted to the FEWELL application.

Step 6: Reassemble the Motorcycle

Once all connections are verified safe, secure, and properly insulated, reinstall the compartment, fuel tank, rear fairing, and seat. Securely tighten all screws and fasteners.

3 Account Registration & Authentication

> **Network Requirement:** FEWELL relies on an active internet connection (Mobile Data or Wi-Fi) for account authentication, user session validation, live route mapping via OSMdroid, and cloud database synchronization.

- **Option A: Email & 6-Digit OTP Sign-In** — Open the app, enter your email address, retrieve the 6-digit verification code sent to your inbox via Gmail Custom SMTP, and enter the code into the app prompt to authenticate with Supabase.
- **Option B: Google One-Tap Sign-In** — Tap *"Continue with Google"* and choose your Google account from the native prompt. (Requires your release APK SHA-1 fingerprint registered in Google Cloud Console matched with your Supabase Web Client ID).

4 Ride Tracking & Telemetry Dashboard

- **Starting a Ride:** Turn on mobile data or Wi-Fi, turn on motorcycle ignition, open FEWELL, confirm the top status indicator displays **ESP32 Connected** (green light), and tap **Start Trip**. Live speed (GPS), live fuel level (%) & remaining liters (INA219 via ESP32), distance (km), and estimated expense update in real time.
- **Ending a Ride & Cost Calculation:** Tap **End Trip**, enter the current gasoline price per liter (e.g., PHP `65.00/L`), and tap **Confirm**. FEWELL computes total trip expense and commits the completed trip record directly to your Supabase cloud database.

5 System Troubleshooting Guide

Issue / Symptom	Corrective Action
"Developer Error" on Google Sign-In	Extract the release SHA-1 key directly from your installed APK using `keytool -printcert -jarfile app-release.apk` and paste that exact SHA-1 fingerprint under your Android Client ID in Google Cloud Console.
OTP Email Failure (500 Error)	Ensure 2-Step Verification is active on your Google Account, generate a 16-character Google App Password, and enter it into Supabase Custom SMTP settings without spaces (Port 587 TLS or Port 465 SSL).
Live Fuel Gauge Bouncing	Verify the ESP32 firmware reads the INA219 current sensor over a moving average window (1–2 seconds) before broadcasting JSON packets over Bluetooth.